a) Yes, it does

b) $\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}$

$S_{xy} = \sum_{i=1}^{15} (x_i - \bar{x})(y_i - \bar{y})$

$S_{xx} = \sum_{i=1}^{15} (x_i - \bar{x})^2$

$\hat{\beta}_0 = \bar{y} - \bar{x}\hat{\beta}_1$

$\bar{x} = 53.2$; $\bar{y} = 42.86$

$S_{xy} = 17024.4$

$S_{xx} = 20586.4$

$\hat{\beta}_1 = 0.826$

$\hat{\beta}_0 = -1.13$

Now, $\boxed{\hat{y} = -1.13 + 0.826x}$

c) Predicted value

if $x = 50$, then

$\hat{y}_{50} = -1.13 + 0.826 \times 50$

$\boxed{\hat{y}_{50} = 40 m^3}$

d) Calcul of standard error deviation

$\tilde{\sigma}^2 = \frac{1}{n-2} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$

$\tilde{\sigma}^2 = \frac{357}{13} = 27.46$

$\tilde{\sigma} = 5.24$

$S_{yy} = \sum_{i=1}^{15} (y_i - \bar{y})^2$

$S_{yy} = 14436$

e) Determination of coeff of determination R^2

$R^2 = r^2 = \frac{S_{xy}^2}{S_{xx} S_{yy}}$

$R^2 = 0.975$

It means that there is a strong linear corelation between y and x.

97.5% of the variation in runoff can be ~~attributted~~ to the linear relationship with rainfall volume.